IS216 - Group Project

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Synopsis

As a group, you are to build a dynamic and interactive web application that utilizes the technologies you've learned this term. This project should include a back-end implementation for data storage and user authentication where necessary. The key learning outcome will be demonstrating your ability to integrate the various technologies you've studied and apply them effectively in your final solution.

Deliverables

1. Progress Pitch
2. Final Project Deliverables
    a. A working web application, submitted via eLearn.
    b. A presentation, recorded and uploaded to **YouTube** (refer to details below).

Progress Pitch

Deadline: *Week 9 (check with your section faculty on the arrangement of progress pitch)*

- Presentation of your project progress
- Demo of your app in progress

Final Project Deliverables Submission

Deadline: *Week 12 Friday 9am (No further changes allowed, Strictly NO extension of deadline)*
The late submission policy is as follows:

Submit **within 1 hour** of the specified deadline	10% penalty (of your score for the entire project)
Submit **beyond 1 hour** of the specified deadline	0 Mark

Zip the following 3 items and submit via **eLearn → Assignments --> Final Project Deliverables**:

- **Presentation Slides/Materials**
    - Your presentation slides and any other materials used for creating your videos.
    - File formats: PPTX or PDF
    - Approximately 10 slides (not a strict requirement).
    - Include a link to the video presentation on the first slide.
        - See the Presentation section below for details on submitting the video recording to YouTube)

- **README**
    - If hosted on cloud: include the URL to the first page of your application.
    - If using public GIT repository: Include the URL to your Git repo.
    - Provide step-by-step instructions on:
        - How to set up your application based on the submitted file(s).

        - How to run your application.
        - How to test your application. Include test cases / scripts if applicable.
    - If there are any username/password details, include them in this file.

- **Application**
    - Your application must be packaged in the root folder (top directory) of your deployed server with the first page accessible via an "index" page.
        - For example, if running locally, "http://localhost:[port]" should load the home page of your app

For the deployed application, you will need to submit the following in addition to the items mentioned above:

- **Deployed application**
    - Submit your app link (*please check with your section faculty or instructor on how to submit the link)*.
    - Ensure that your app is running and remains operational. Do not restart unless we contact you for further instructions.

IMPORTANT: You can safely stop, delete, or disable your app, instance, API keys and accounts (especially those linked to your credit cards) after Week 18.

Minimum Requirements

Working application

- ❑ You must identify and solve a problem. This can be either an unsolved problem or an existing problem. (e.g. a booking system, contact tracing, escape room game, etc.).
- ❑ You must use HTML, CSS, and JavaScript in your implementation. Optionally, you are suggested to use Bootstrap (or similar CSS framework), JavaScript, JSON, and Vue (or a similar JavaScript framework).
- ❑ The solution must include a responsive design. Your UI design should follow the recommended UI design patterns you learnt from other modules such as IDP.
- ❑ The solution must be a dynamic site that interacts with a backend data store. This could involve applying knowledge from IS113 (e.g. Express.js + MongoDB), but you are free to choose another data storage solution if you prefer (e.g. files, other types of databases, cloud storage).
- ❑ Your solution must make meaningful use of at least one external publicly available API via asynchronous HTTP requests.
- ❑ Your solution should include (at least End-to-End) Testing of the application's core features.

Presentation

- ❑ Record your presentation **(maximum 12 minutes, at least 720p - ideally 1080p)** and upload it to YouTube. Please ensure that the video is either unlisted or public.
- ❑ Present and demonstrate your application within the specified time limit.
- ❑ Focus on the following:
- ❑ On the Solution
    - What problem is your group trying to solve (e.g. escape room), and how is solving this problem useful?

- Briefly describe how you solved the problem and met the project requirements.
  - Discuss the design of your solution.
  - Explain how you used JavaScript, Vue, or similar JavaScript frameworks to make your app dynamic and interactive.
  - Describe how you used CSS, Bootstrap, or similar CSS frameworks to style your app, ensuring that the web pages are responsive, clean (no irrelevant or extraneous information), and have clear, logical flows (intuitive to use).
- Clearly state the type(s) of data store(s) you are using.
- Identify the external APIs you have used and provide brief integration details.
- Describe how you test your application's features.

❑ On the Presentation

- Ensure good flow and transitions.
- Aim for ease of understanding.
- Use concise and meaningful presentation materials (slides, videos, etc.).
- Manage your time well:
  - Having a lot of extra time left with a presentation still unclear is not ideal.

Components

Progress Pitch (5 marks)

Component *(Check with your faculty/instructor how the progress pitching will be conducted)*		%
	Clearly defined problems & features	25%
	Clearly defined web design & tech stack	25%
	Clearly defined job scopes / task delegation	25%
	A functioning task – e.g., backend service / API call / CRUD operation – a feature or a sub-feature	25%

Final Presentation and Deliverables (20 marks)

Component		%
(1) Presentation		**25%**
	The presentation	20%
	Q & A	5%
(2) Working Web Application		**75%**
	Problem(s) solving and solution appropriateness	20%
	Working app, correctly implemented and usable solution	27%
	Styling (including responsive design)	18%
	Testing	10%

Peer evaluation (5 marks)

Each team member will submit the peer evaluation form, which will be evaluated based on the teaching team's professional judgement. For any conflict, the team should first try to manage and resolve internally rather than simply waiting it out and stating the conflicts in the peer evaluation form. If a member is difficult to manage, report to the teaching team early (**at least 3 weeks before the deadline**) so that the teaching team can intervene and take action. If the teaching team judges that the team did not make such an effort to manage conflicts, all the team members may still get the same grade, regardless of peer evaluation scores. Teamwork and project management is an implicit part of the project grading.

In an exceptional situation where a member is not contributing anything significant or s/he is missing majority of the team meetings/discussions, the member will be penalized depending on the severity, including the possibility of zero mark for the entire project.

Marking Rubrics

Progress Pitch

Marks	Details
C+ & below	Problems / features / job scopes / web design / tech stacks are vague. There is no functioning task.
B- to B+	Problems / features / job scopes / web design / tech stacks are not very clear but somewhat defined. A task is somewhat functioning.
A- to A	Problems / features / job scopes / web design / tech stacks are clearly defined. A task is functioning well.
A+	*Challenging* Problems / features / job scopes / web design / tech stacks are clearly defined. A *challenging* task is functioning well.

Presentation

Marks	Details
C+ & below	**Presentation with Demo sloppy.** ❑ Difficult to understand how the solution addresses the problem, or the clarity of the technologies used.
B- to B+	**Presentation with Demo is of reasonable standard.** ❑ Relatively clear presentation and demo that shows how technologies are used to address the problem with some minor hiccups.
A- to A	**Presentation with Demo is good.** ❑ Clear presentation and demo that are quite seamless. Easy to understand, with a clear explanation of what was done and how it was accomplished.
A+	**Presentation with Demo is exceptional.** ❑ Crystal-clear presentation and demo that are seamless. No clarification questions are needed to understand what was done. Additionally, at least one bonus point must be earned to qualify.

Deliverables

Problem(s) solving and solution appropriateness

Marks	Details
C+ & below	**Problem is not solved and/or not a useful app.**
B- to B+	**Problem is somewhat solved and is relatively useful.** ❏ The problem might not be completely solved but is mostly addressed.
A- to A	**Problem solved is very useful and is challenging** ❏ A clearly defined problem that is addressed with a satisfactory solution.
A+	**A highly challenging problem is solved and the solution is excellent.** ❏ It is a challenging problem to address, and the solution is highly appropriate.

Working app, correctly implemented and usable solution[1]

Marks	Details
C+ & below	**App is somewhat working to not working.** ❏ Intermittently working, with some parts functioning or completely non-functional.
B- to B+	**App is buggy with some working parts.** ❏ Contains bugs, but none are critical or show-stopping. The app is roughly usable without requiring excessive effort to understand how to use it.
A- to A	**App is working well, and usability is good.** ❏ Some minor bugs, but the app is relatively intuitive to use.
A+	**App is working perfectly with great usability.** ❏ No bugs at all; the app is completely efficient and intuitive, requiring no effort to figure out how to use it.

Styling (including responsive design)

Marks	Details
C+ & below	**Limited styling implemented, minimal responsiveness.**
B- to B+	**Relatively decent styling and somewhat responsive site** ❏ Somewhat clean and appealing, and the site is somewhat responsive, with minor issues but nothing critical.

[1] In the sense of the following guidelines: https://www.interaction-design.org/literature/article/user-interface-design-guidelines-10-rules-of-thumb

A- to A	**Good style and somewhat responsive site** ❑ Attractive and well-executed, and the site is fully responsive.
A+	**Excellent style and fully responsive site** ❑ A beautiful site that makes effective use of style and is fully responsive, scaling appropriately for different screen sizes and orientations.

Testing

Marks	Details
C+ & below	**Limited coverage of core features.** ❑ Tests are limited, generally non-repeatable and not stable. Provide very limited instructions in README on how to run tests.
B- to B+	**Decent coverage of core features** ❑ Tests cover some of the app's main user journeys. Some of the tests are repeatable and stable. Provide some brief instructions in README on how to run tests. Tests demonstrate somewhat understanding of testing concepts.
A- to A	**Good coverage of core features with some quality test cases** ❑ Tests cover most of the app's main user journeys. Tests are mostly repeatable, generally stable and have clear structure. Provide somewhat clear instructions in README on how to run tests. Tests demonstrate a good understanding of testing concepts.
A+	**Excellent coverage of core features with quality test cases** ❑ Tests cover the app's main user journeys. Tests are repeatable and use stable selectors and have clear test structure. Provide clear instructions in README on how to run tests. Tests demonstrate excellent understanding of testing concepts.

Frequently Asked Questions (FAQs)

Last updated: **10th August 2025**

1. Are we required to use Bootstrap?

No, you are not required to use Bootstrap specifically. You may use Bootstrap or any similar CSS framework or just vanilla CSS. However, using a CSS framework is encouraged.

2. Are we required to use Vue.js?

No, you are not required to use Vue.js specifically. You may use Vue.js or any similar JavaScript framework or just vanilla JavaScript. However, using a JavaScript framework is encouraged.

3. I want to use React. Is this allowed?

Yes using React or any similar framework is allowed. However, we encourage you to use Vue.js for two reasons: (1) we can provide better technical support since Vue.js is covered in the syllabus, whereas React is not; and (2) using Vue.js will help you prepare for the final exam, which will include questions on JavaScript and Vue.js, but not on React. Also, consider your teammates' familiarity with React. If everyone is well-versed in React, it's fine. However, if some teammates are not, they would need to learn two different JavaScript frameworks by the end of IS216. Ensure that they are comfortable with this.

4. Can my instructional team provide technical support on the technology we are using, e.g., Docker, AWS, etc.?

In general, no. There is a plethora of technologies out there. Given the variability in our expertise with different technologies, it would be unfair for us to assist one team with technology we are familiar with while being unable to support another team with technology, we are not knowledgeable about. However, we are happy to engage in high-level discussions regarding the appropriateness of different technologies for your solution.

5. Which browser should we use?

You can use any browser you prefer. However, for grading purposes, please ensure that your website displays properly in Google Chrome. We will also use Chrome's developer tools to test your site's responsive design.

6. Are we allowed to use publicly available code sources (e.g. from GitHub projects)?

You may use **publicly and freely available** libraries (i.e., not paid). However, such usage must be clearly identified in your presentation and within the comments of your code. You must **NOT** present this code as your own work, as that would constitute plagiarism. If you are unsure, please contact your instructional team to discuss your plans.

7. Are all members of a group receiving the same group project grade?

Yes, a single grade is awarded for the project. However, if a group member is not participating, please inform your instructors *EARLY*, and individual adjustments will be considered on a case-by-case basis. But if you inform your instructors very late (one week before your final deliverables), it is considered too late to take any meaningful action and your instructors may reject your case.

8. Are all members of a group required to code?

Yes, all members should contribute to the coding of the project. It is advisable to assign specific tasks to each member (e.g., Bob handles styling, and Olivia manages scripting) to ensure effective collaboration.

9. Are all members of a group required to present (i.e. in the YouTube video)?

Please check with the instructional team for your specific section to confirm the requirements.

10. Are we allowed to use back-end code from IS113 and build on it?

Yes, you are allowed to build upon example back-end code from IS113. In IS216, we will provide an example of how to apply styling and scripting to some back-end code, and you are welcome to build on that as well if you wish.

11. Will you be grading the backend code?

While there is no explicit grading component specifically for the back-end code, the overall functionality of your application is evaluated. Therefore, the quality of your back-end implementation will be assessed under the grading criterion for a “working app, correctly implemented, and usable solution.”

12. What is the best way for our team to collaborate on source code together?

Please discuss with your instructional team if you need recommendations on tools and strategies for effective source code management and collaboration.

13. Is the usage of dummy data discouraged?

If the type of data your application requires is publicly available, we encourage you to use real data to make your prototype more realistic. However, if you anticipate difficulties in obtaining real data, please consult with your instructional team for guidance.

14. Under “problem solving and solution appropriateness”, are you referring to coding complexity, or more to the business side?

We are referring to the business side, meaning you should clearly articulate the problem your website addresses and demonstrate that the implementation effectively solves this problem. The problem does not need to be a critical societal issue. For example, a virtual escape room may not address a major societal problem but still solves a relevant issue, such as enabling remote connection with friends and family, especially during a pandemic.

15. If other groups are tackling the same problem, would we be penalized?

No. There are often many ways to solve the same problem, and multiple groups working on similar issues is acceptable.

16. Are we allowed to scrape data from a website?

If you choose to scrape data from a website, you must ensure that you comply with any restrictions outlined in the site's robots.txt file and consider any legal implications. Additionally, you must clearly identify and properly credit the sources of your data. Note that using scraped data does not substitute the requirement to utilize at least one publicly available API.

17. How will the responsive design of our web application be tested?

We will test your website using Chrome's developer tools across various viewports, starting from the iPhone 6 to the XL breakpoint of Bootstrap. Ensure that your layout looks good and adjusts properly across all viewports within this range.

18. Can we see an example of a past project to gauge how much functionality is needed?

Yes. Your instructional team will demonstrate a successful project from the previous year in Week 5 or 6 to help you understand the level of functionality we expect.

19. The API I'm using is free but has usage limits. How much testing will be done after submission?

As a rough estimate, each functionality will be tested up to 5-6 times per day. You will need to assess whether this falls within the free usage limits of your API.

20. Are we allowed to use AI/LLM in the project?

Generally, we allow the use of AI/LLM for lower order thinking tasks such as

- information search
- Generating website concepts, layouts, or themes
- Exploring different UI/UX design inspirations
- Explaining coding errors and debugging hints
- Boilerplate code generation – starter code and code snippets
- Generating unit test cases, sample inputs, or mock data

You are not allowed to use AI/LLM for higher order thinking tasks such as

- Core implementation tasks (to implement main business logic, backend endpoints, or critical frontend interactivity)
- Problem solving – students must demonstrate their own capability for solving problems and addressing major implementation issues, not just rely on automated AI fixes.

In the README.txt file, state clearly the parts you solely rely on AI/LLM.